Takeoffs and Test Scores: The Impact of Noise Pollution on Standardized Test Performance

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Abstract

Noise pollution is ubiquitous in urban life, but little is known about its effects on educational attainment. I estimate the causal effect of noise pollution on standardized test performance by isolating exogenous variation in aircraft noise pollution and implementing a value-added model to account for differences in student ability. I find a large negative impact on math test performance, but no impact on English Language Art test performance. Moving from a school at the 25th percentile of exposure to one at the 75th percentile is associated with a math test score reduction of 0.1 standard deviations, twice as large as the impact of missing 10 school days during the year. I find similar results using a second identification strategy that leverages changes in aircraft noise from the introduction of new flight paths. These findings suggest a potentially large value to investments in noise insulation for schools exposed to high levels of noise pollution.

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1 Introduction

Noise pollution, unwanted sound from transportation, construction, and other activity, has become a feature of modern urban life. High noise levels, even those well below the level required for hearing damage, are linked to increased stress, poor sleep, and adverse cardiovascular health outcomes (Münzel et al. 2014; Saucy et al. 2020; Vienneau et al. 2022; Evans et al. 2001), resulting in a significant public health burden (World Health Organization. Regional Office for Europe 2011). Even at relatively low levels, noise has negative non-health effects in the form of reduced productivity (Noweir 1984; Dean 2024). These negative impacts are especially concerning given how pervasive noise pollution is in everyday life. In the United States, more than 45 million people live within 300 feet of major transportation infrastructure such as a large road (Environmental Protection Agency 2014) and one in eleven public K-12 schools is within 500 feet (Hopkins 2017).

Despite the ubiquity of noise in modern life, there is only a nascent literature on the causal impacts of noise on cognition and learning. Recent work has found that increased noise impairs memory and reduces productivity (Hygge 2003; Jahncke et al. 2011; Dean 2024). Noise insulation on building has the potential to reduce impacts, but the prevalence of noise may lead individuals to underestimate its effects. Prior research suggests workers are unaware that increased noise reduces their productivity and therefore do not make valuable defensive investments (Dean 2024). Therefore, documenting the effects of noise pollution is important to ensuring schools invest in the efficient amount of noise insulation and other defensive technologies that can mitigate negative impacts of noise pollution.

Cognitive psychologists have documented correlations between noise and student performance and comprehension (Evans et al. 1995; Evans and Maxwell 1997; Haines et al. 2001; Stansfeld et al. 2005; Matheson et al. 2010; Klatte et al. 2013).[1] However, these cross-sectional studies do not account for residential sorting. There is a well documented property value penalty associated with noise pollution such that houses in louder neighborhoods tend to have lower property values (Crowley 1973; Nelson 1980 2004; Chang and Kim 2013). Because most students attend local schools, students attending schools exposed to more noise pollution may be from lower income households than those attending schools in quieter areas. Given these results, researchers should worry estimates relying on cross-sectional variation in noise likely capture both direct impacts from noise pollution on cognition as well as indirect impacts from residential sorting. In this paper, I leverage changes in aircraft noise pollution at schools around Seattle-Tacoma International Airport

[1] There are a handful or experimental studies that find impacts of noise on memory tasks (Hygge 2003; Hygge et al. 2003; Jahncke et al. 2011). These studies expose older, usually college-aged students to relatively low-level noise while reading and attempting to remember what they read. These studies unfortunately do not measure any impacts on standard measures of learning and do not attempt to determine whether noise impacts non-reading recall or skills. Potentially due to their often small sample sizes or participant motivation, results in this field have contradicted each other (van Kempen et al. 2010).

(SeaTac) to estimate the causal impact of noise pollution on standardized test performance.

Utilizing data from noise monitors around SeaTac, I measure noise from overhead airplanes near schools from 2009 through 2019. I implement a value-added model of student performance at schools near monitors to adjust for student ability and isolate the direct effect of noise pollution on performance. My approach exploits both across- and within-school variation in noise pollution over time that comes from a change in flight paths into and out of SeaTac as well as year to year variation in flight noise. My results show that noise pollution had a negative effect on math test performance but no impact on English Language Arts test performance.

Importantly, the effect size of the impact on math test scores is large. Under my preferred specification, I estimate that a one decibel increase in aircraft noise during the school year is associated with a 0.0070 standard deviation decrease in math test scores. These are large effect sizes. Moving from a school in my dataset at the 25th percentile of exposure to the 75th in 2019 is associated with a 0.10 standard deviation reduction in math test scores, twice as large as the impact of missing 10 days of school (Liu et al. 2021).

My work primarily contributes to the literature on identifying causal impacts of noise pollution in the labor context. Dean (2024) uses a randomized control trial at a textile training facility in Kenya to document a negative impact of noise on cognitive task productivity. The study also elicits willingness to pay for quiet working conditions and finds that workers who are paid based on output are no more willing to pay for quiet conditions, suggesting workers are unaware that increased noise reduces their productivity. Hener (2022) uses random variation in which direction planes approach Frankfurt airport on a given day that comes from wind direction to estimate that higher daily background noise increases the violent crime rate. I identify a causal impact of noise pollution using year to year variation in flight noise. I also use a second identification strategy that takes advantage of a policy that changed landing and takeoff paths. I am not the first to use this policy change; Kuminoff and Mandia (2025) and Allroggen et al. (2025) leverage the same shift to estimate the effect of noise pollution on the Phoenix and Seattle, Chicago, and Boston housing markets, respectively. Beyond my results, I add to this literature by showing that there is a large amount of year to year variation in flight noise around Seattle Tacoma International Airport before and after the flight path change that can be exploited to measure noise pollution impacts.

This paper also contributes to the broader and more robust literature measuring the causal impacts of pollution on education and human capital accumulation. It is well documented that both contemporaneous (Ebenstein et al. 2016; Graff Zivin et al. 2020; Zhang et al. 2018) and cumulative (Heissel et al. 2022; Wen and Burke 2022; Stafford 2015) air pollution exposure reduces exam scores. Beyond diminishing test scores, air pollution is tied to increased absence rates and behavioral incidents (Heissel et al. 2022). Cumulative lead pollution also diminishes student achievement (Hollingsworth et al. 2025) and increases absences and suspensions (Gazze et al. 2024). This paper

extends the work on the adverse impacts of pollution exposure on educational outcomes to include noise pollution.

The remainder of this paper is organized as follows. Section 2 provides information on noise levels and the implementation of new flight paths at SeaTac. Section 3 discusses noise data and the construction of an analytic dataset. Section 4 explains two identification strategies for estimating the causal impact of loud flights and flight noise on test performance. Section 5 presents results from these models. Finally, Section 6 concludes.

2 Context: Performance-Based Navigation

Over the past two decades, the Federal Aviation Administration (FAA) has made a series of changes to commercial flight paths and procedures. The backbone of this suite of programs commonly known as NextGen is an advancement in navigation technology. Starting in the early 2010s, the FAA began using Automatic Dependent Surveillance-Broadcast (ADS-B) technology to track planes and mandated all planes be equipped with ADS-B by 2020. ADS-B determines an airplane's position using GPS (satellites) rather than radar (radio waves), which allows for more precise location tracking. This advancement allowed the FAA to shift more planes to the most efficient landing and takeoff routes to reduce approach and takeoff flight distance.

Using ADS-B, the FAA implemented Performance-Based Navigation (PBN), a program that altered the routes pilots used for takeoff and landing, at the nation's busiest airports. The program sets limits on flight paths, giving air traffic controllers and pilots less autonomy in determining routes. With more precise location information from ADS-B, air traffic controllers can safely have planes fly in closer proximity rather than encouraging pilots to spread out over a wider area. This has concentrated more planes on the most efficient routes for takeoff and landing.

PBN has caused noise corridors to tighten and become louder as planes fly over the same optimal paths in quick succession, leading to pushback from members of the public. Airports and the FAA have seen dramatic increases in noise complaints (Snow and Lowy 2017).[2] When flight paths shift, there are both winners and losers in terms of noise pollution – some areas experience declines and others increases. This project does not attempt to measure the impact on total social surplus, but to instead estimate the effect on a subset of those who were affected by the path changes.

The FAA piloted new PBN flight paths on 1% of flights at SeaTac beginning in March 2013 (United States Department of Transportation 2014). SeaTac expanded the program to all planes equipped with ADS-B technology in April 2015 (Alaska Airlines 2015). As a result, flight paths

[2]These complaints do not impact the way that air traffic controllers route flights day to day, though a lawsuit in Phoenix did lead to temporary reversion to the previous flight paths.

used for landings and departures concentrated along more narrow corridors. PBN induced both immediate and gradual changes in flight path use because ADS-B technology was not required until 2020. In Section 4.1, I show that the schools in my data were exposed to large year to year fluctuations in noise rather throughout the sample period than a large shock and steady increase.

3 Data and Treatment Measurement

3.1 Noise Data

The Port of Seattle operates a network of noise monitors around SeaTac, and 24 of their monitors covered 2009 through 2019. These monitors record sound pressure continuously and document all time spans that correspond to an airplane flying overhead. The Port of Seattle quantifies the sound pressure from flights with a commonly used measure, sound exposure level (SEL). My data capture each instance a flight overhead results in the SEL exceeding 65 decibels (dB), approximately the same volume one experiences using a vacuum cleaner or standing next to city traffic. The vast majority of the events captured by the monitors are not loud enough to cause hearing damage, but they are loud enough to startle, distract, or drown out talking (Online Appendix Figure 1 provides the distribution of loud noise events).

SEL events are helpful for counting the number of loud noise events, but not the amount of noise students hear. I opt to use a common measure of aggregate noise, the equivalent continuous sound level (L_{eq} or, more commonly, Leq), to measure the noise students experience. Leq measures the total noise pressure from loud noise events during a period of time rather than the frequency of loud noise. The decibel scale is logarithmic, so a 90 dB noise generates 100 times as much sound pressure as a 70 dB noise (even though the human ear only perceives the louder event as four times as loud as the quieter). This means that the loudest events largely determine the Leq.[3] The Port of Seattle collects Leq data and calculates a level for both community and aircraft noise, which allows me to isolate noise from aircraft from community noise, which may be endogenous to economic conditions.

In this paper, I use aircraft-specific Leq and the number of SEL events as treatments. The school year runs from the start of September to the start of June in Washington, and schools are allowed to administer standardized tests any time during the final 12 weeks of the school year. I do not have data on the date of exams and therefore cannot determine the amount of exposure during exams. I calculate the average number of SEL events per hour and aircraft Leq from September through

[3]For example, consider one hour at two different noise monitors: at one there was one flight overhead with a SEL of 90 dB, at the other there were 100 flights overhead each with SELs of 70 dB (and no other flight noise besides these flights). The aircraft-specific Leq of these two monitor-hours is the same even though one indicates potentially 100 times as many interruptions.

the end of February to focus on the effect of cumulative exposure during learning. However, noise pollution during March, April, and May is highly correlated with noise pollution during the rest of the school year, so I cannot separate the impact of noise pollution during exams on recall from the impact of cumulative noise pollution on learning.[4]

One difficulty with the SEL data is determining when data is missing. One quarter of hours in my sample have no SEL events. This could reflect that there were no loud flights overhead during that hour or that the monitor was not collecting data. The probability of not recording any SEL events is non-random both across monitors and days. For example, the share of monitor-days with no SEL events is lowest in December when there are more flights. To address occasions where data is missing across all stations, I drop all days where at least three quarters of stations do not record a single SEL event.[5] These days are concentrated in five school year months (November and December 2010 and January, February, and November 2015) which are missing data. Of the 115 school days I classify as missing data, 108 are during these five months. Unfortunately, most monitors are missing all data for the first four months of 2015. Without data for the start of the year and April, when flight paths changed, I cannot tell whether flight volumes changed notably between the learning period and exam period in 2015. I choose to drop the 2014/15 school year from my data because if there is a large increase during the exam period in 2015 but I record a school year with no large change in cumulative exposure, then including the school year would increase measurement error. I conduct sensitivity analyses including the 2015 tests and show that this choice does not change my results.

To address cases where specific monitors are not collecting data, I count consecutive days with no SEL events. I drop all hours during spells of at least two weeks without an SEL event. This process removes 3.6% of remaining monitor-hours. The two-week threshold trades off potentially underestimating the number of SEL events (by including hours when monitors aren't functioning) with potentially overestimating (by excluding hours where there were no flights overhead for many days in a row). Changing the threshold by a week in either direction has a very small impact on the share of hours dropped and no impact on the results.

3.2 Student Data

Washington state's Education Research and Data Center provided data on student enrollment, demographics, and test scores for this project (Education Research and Data Center 2025). My outcomes of interest are annual performance on mathematics and English Language Arts (ELA)

[4]The correlation between Leq from September through February and March through May at the noise monitor by year level is 0.96.

[5]This is to remove the 5% of days with a high missing rate, which are usually days where all monitors or around 80% of monitors are missing data. Only 1.2% of days have missing rates between 50% and 75%, so the decision to set the threshold at 75% is not impactful.

standardized tests taken by all third through eighth grade students. I measure a student's performance on a test as number of standard deviations from the statewide mean that year. This accounts for year to year changes to the testing instrument that impact difficulty.

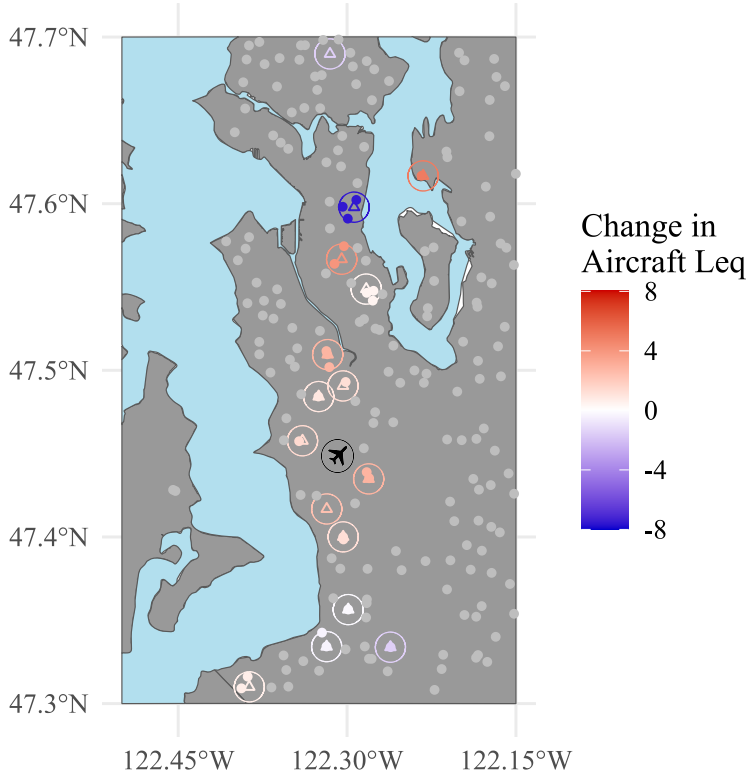


Figure 1: Monitor and School Matching

The Washington Office of Superintendent of Public Instruction (OSPI) provides data on school locations. I match monitors to all schools within one kilometer.[6] Figure 1 provides a visual representation of this process. Each monitor is depicted as a shape at the center of a circle with a 1km radius. The shape of each monitor shows the difference in aircraft Leq between the post- and pre-PBN periods (the 2016-2019 and 2010-2014 school years, respectively). Schools inside the 1km radius around a monitor are matched and depicted in black rather than the gray of the unmatched schools. My final dataset includes school years for all third through eighth grade public school students who were enrolled at a single school during the school year.

[6]Increasing the radius around monitors increases the number of matched schools and reduces standard errors, but also each marginal monitor is exposed to noise less similar to the monitor, increasing measurement error and attenuating estimates. In Section 5.2, I show that my results are not sensitive to this choice.

4 Empirical Design

4.1 Variation in Noise Pollution Exposure

In my setting, treatment is assigned at the school year level and there is no variation in treatment exposure within schools during a school year.[7] Variation in noise pollution comes from assignment to schools, year to year variation in flight path use, gradual increases in the number of flights in and out of SeaTac, and the shift in flight paths from the PBN implementation. If noise pollution were randomly assigned to schools in all years, one could simply regress test scores on noise exposure to recover a causal estimate of the effect of noise pollution. However, noise pollution is consistently higher at some schools than others, which could lead to residential sorting resulting in different types of students at more and less exposed schools. Therefore, the key challenge to identifying a causal impact is ensuring that the noise variation I isolate is not correlated with student characteristics that explain test scores.

The most common tool for assessing the impact of a one-time policy change is the difference in differences model. In this context, a difference in differences model with continuous treatment can compare the outcomes for students before and after the flight path shift and would attribute differences in these differences to the relative change in aircraft noise. This approach captures variation in noise from flight path changes both from an immediate shock and the average of gradual increases or decreases following the changes. The difference in differences model however does not leverage year to year variation in noise pollution or student assignment to schools.

To understand the relevant sources of variation I observe in my data, I plot year to year change in noise pollution against prior year annual noise pollution in Figure 2 (I show annual flight noise Leq by monitor in Online Appendix Figure 2). In the figure, each point represents a monitor-year, the color denotes the monitor, and the shape denotes the year relative to the PBN implementation. If the PBN implementation had an immediate, permanent effect on noise pollution at each monitor, then all points denoting other years (circles and squares) would be clustered tightly around the X-axis because there would be very little year to year change,[8] and points for the year after the shift (triangles) would capture the vast majority of noise variation. If the implementation induced both an immediate shock and gradual shift, then pre-implementation years (circles) would be clustered around the X-axis with post-implementation years (triangles and squares) all on one side of the X-axis for each monitor. The figure instead shows a mix of shapes on each side of the X-axis for most monitors, suggesting that year to year variation is a key component of the noise pollution schools experience.

[7] This is a limitation of the data. It is possible that some students take classes in rooms that are more exposed than their peers, however I have no information on student classrooms or school acoustics.

[8] There was growth in flight traffic in and out of SeaTac, so both circles and squares would likely be slightly positive.

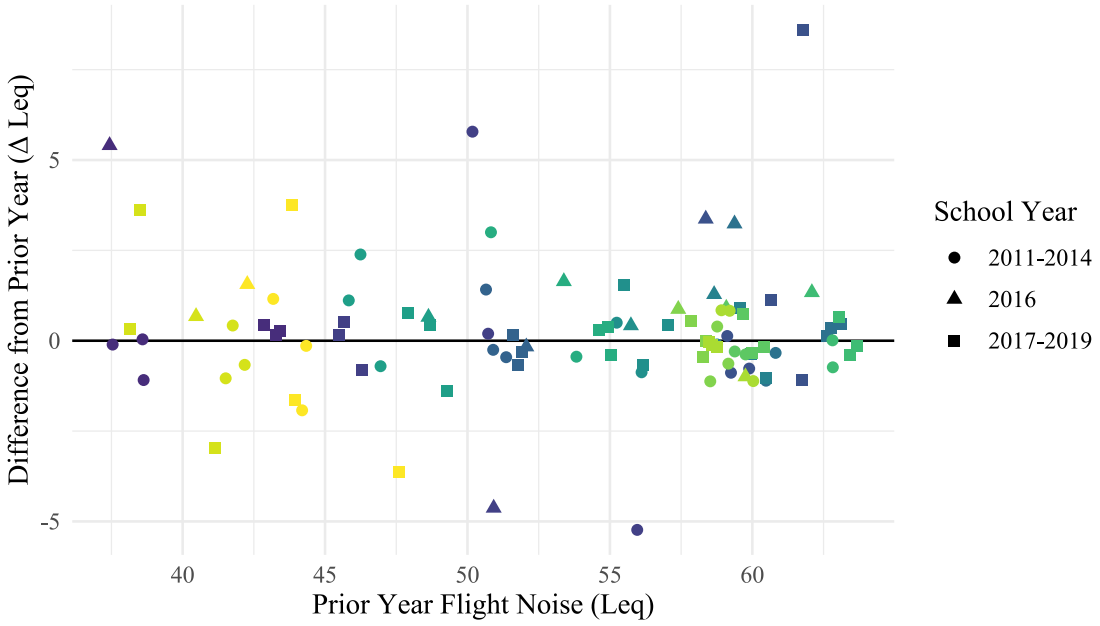


Figure 2: Annual Change in Flight Noise by Prior Year Flight Noise

I utilize student data to capture the year to year variation in noise pollution while still accounting for potential sorting. I also estimate a continuous treatment difference in differences model to show that my results are not driven by correlation between noise and unobserved differences in student ability (after controlling for prior year performance). Additionally, I use the difference in difference model to estimate the effect of the PBN implementation on student demographics and show that it does not appear to have changed sorting patterns. I describe this model in Section 4.4

4.2 Value-Added Model

I identify the impact of noise pollution on student performance by implementing a value-added model. The logic of the value-added model is that students perform similarly across years, so past test scores are a good predictor of future test scores. I model student i's standardized test performance as a function of the student and their school, $s(i)$. Specifically, it is a function of the number of loud noise events at school $s(i)$ during school year t, a vector of student characteristics, X_{it}, that includes student demographics and their test performance from the prior year, and idiosyncratic variation in performance, ε_{it}. To account for variation across years in test difficulty, I measure performance, y_{it}, as the number of standard deviations from the mean score for all Washington students in grade g during year t. I estimate the following equation using ordinary least-squares (OLS) regression:

$$y_{it} = \beta X_{it} + \gamma FlightNoise_{s(i)t} + \alpha_1 PM_{s(i)t}^{learning} + \alpha_2 PM_{s(i)t}^{testing} + \lambda_{s(i)g} + \upsilon_t + \varepsilon_{it}. \quad (1)$$

The parameter of interest is γ, which is the estimate of the causal impact of increased noise, either a 1 dB increase in aircraft Leq or an additional loud noise event per hour during a school year. The vector X_{it} includes the student's lagged test performance, gender, race/ethnicity, whether their primarily language spoken at home is not English, and whether they receive free or reduced-price lunch during the school year. I control for average daily $PM_{2.5}$ during the learning period of the school year (September-February) and the testing period (March-May) separately.[9] $\lambda_{s(i)g}$ are time-invariant school-grade fixed effects that account for any differences in school composition or quality not captured by demographics, such as teacher quality or school resources. I standardize test scores relative to all students in the state, so I include school year fixed effects, υ_t, in case performance of Seattle-Tacoma area schools relative to the rest of the state) varies over time.

I use the value-added model to estimate the effect of year to year variation in noise pollution on student test performance. The identifying assumption is that noise pollution exposure is uncorrelated with student ability after controlling for the student's prior year exam performance, school and grade at that school, and student demographics. By controlling for student ability using prior year exam performance, this model and student-level data can mitigate concerns about a negative estimate being the result of sorting rather than noise impairing student learning or test taking.

4.3 Smarter Balanced Assessment Implementation

One complication with my context is that Washington switched from testing students using the Measurements of Students Progress (MSP) to the Smarter Balanced assessment system (SBA) during the 2014/15 school year. This means that a large shift in noise pollution coincided with a change in test instrument. The left panel of Figure 3 shows that students were less likely to meet the exam standard after the switch to the SBA than they were before.

The test regime change creates four problems for this analysis. First, students are less likely to meet the exam standard on the new exam and the tests have different scoring scales. I use standard deviations from the state mean score to scale everyone's performance so that this impact does not affect the results.[10] Second, prior to 2014/15, all third through eighth grade students took a reading exam and both fourth and seventh grade students also took a writing exam. The SBA assessments

[9] After adjusting for the school and year averages, both measures of flight noise are uncorrelated with the average daily $PM_{2.5}$ during both the learning and testing period of the school year.

[10] One exception would be if the change in test instrument had differential impacts across students of different demographics. I show in Online Appendix Figure 3 that there do not appear to be notable differences in impacts across demographic groups.

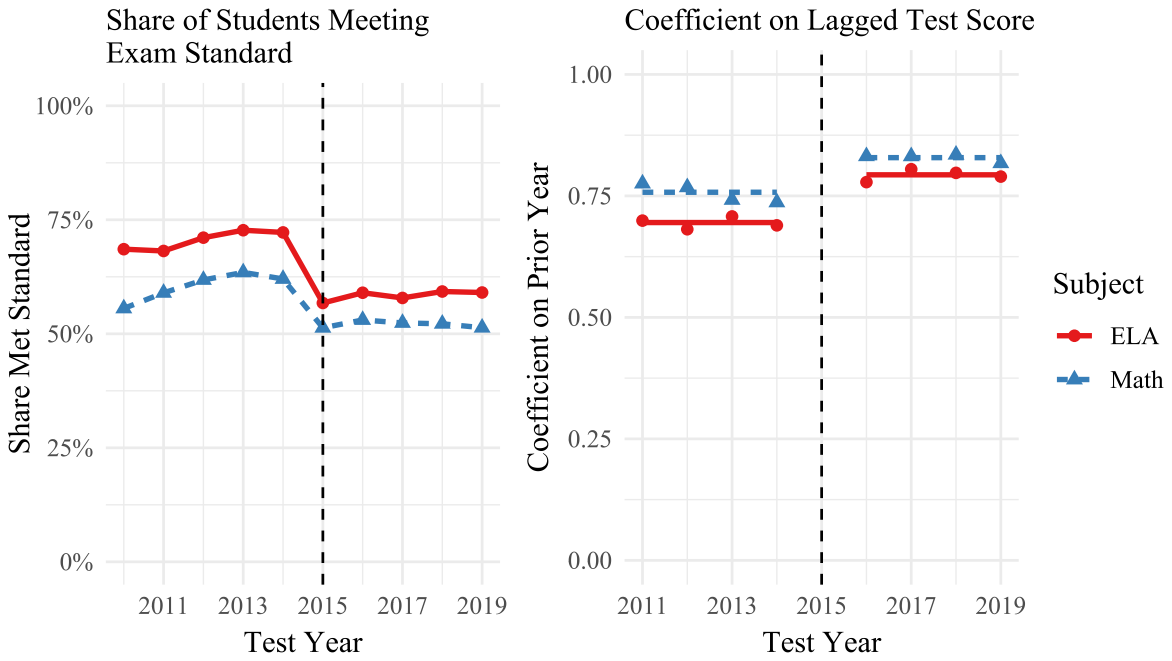


Figure 3: Impact of Test Regime Change

used from 2015 onwards are composed of just a single ELA exam rather than multiple tests. To minimize the impact of this change, I use the reading exam only prior to 2015 and the ELA exam from 2015 onward. Third, during the final year of the MSP regime, schools could choose whether or not to administer the old exam, resulting in a smaller sample that year.[11] The value-added model I use addresses concerns about selection into test taking by conditioning on prior year performance. I also conduct a sensitivity analysis around the inclusion of this year and find that excluding it does not change my results.

The final concern pertains to the prior year test score variable in the value-added model. Lagged test scores serve as a measure of student ability, but if their predictive value varies across testing regimes, then residuals will be correlated with testing regime. This is particularly concerning because the testing regime and flight paths change during the same year. The right panel of Figure 3 shows that the SBA exams better predict subsequent exam performance than the MSP exams do. In this figure, each point is the coefficient on the previous year's exam performance from a regression of performance on prior year performance interacted with test year. The horizontal lines are the coefficients from a separate regression where prior year performance is interacted with assessment regime instead of year. The annual and assessment regime coefficients show that within testing regime, each year's testing instrument is similarly predictive of future performance.

In my model, I interact prior year performance with the assessment regime to account for

[11] Schools did not have the option to implement the new exam, just the choice on whether or not to implement any exam.

the difference in predictiveness of testing instruments. This solves the problem of using a single variable for lagged performance, which would overestimate the prediction value of the prior year performance variable under the MSP regime and underestimate under the SBA regime, potentially leading to more residual variation to explain during the years that noise pollution levels across schools diverge slightly more. One might have additional concerns that the more predictive previous year performance is, the less residual variation there is for noise pollution to explain. The SBA testing instrument is more predictive than the MSP instrument and was used when variation in noise pollution increased following PBN. Therefore, this would bias my results towards zero, leading me to underestimate rather than overestimate the true effect of noise pollution.

4.4 Difference in Differences Model

I estimate a difference in differences model with continuous treatment that leverages the PBN implementation. This model captures variation in school-level noise exposure differences between before and after SeaTac implemented the PBN flight paths in 2015. I estimate the equation:

$$\begin{aligned} y_{it} = &\alpha_0 \Delta FlightNoise_{s(i)} + \alpha_1 post_t + \beta[\Delta FlightNoise_{s(i)} \times post_t] \\ &+ \mu_{it} + \rho PM_{s(i)t} + \lambda_{s(i)g} + \varepsilon_{it} \end{aligned} \tag{2}$$

where $\Delta FlightNoise_{s(i)}$ is the difference between the 2016-2019 and 2010-2014 average school year aircraft Leq and $post_t$ is an indicator variable for whether the school year was 2016 or later. The parameter of interest is the difference in differences estimator β. I include the same vector of student demographic variables, now represented as μ_{it} to denote the exclusion of the prior year test performance variable.[12] I also consider versions of the model using the change in number of SEL events instead of change in average Leq. As in Equation 1, I include school fixed effects.

There are two important threats to causal identification under this design. The first is whether the parallel trends assumption – that in the absence of PBN implementation, differences in test scores between more and less exposed schools would have remained the same – is valid. I consider event study versions of the model comparing all school years to the 2014 school year to show that there are no trends during the 2010 to 2014 school years that suggest that the difference between these schools would not have remained at the same level during the 2016 through 2019 school years. These results are available in Online Appendix Figure 4 and Online Appendix Figure 5. The second concern is that the change in flight noise induces families to sort differently than before the PBN implementation in such a way that more higher achieving students move to less exposed

[12]Noise variation is no longer assumed to be at the school year level, so the prior year test performance variable captures both student ability and treatment.

schools or more lower achieving students move to more exposed schools. In Online Appendix Table 1, I show that there are no significant differences in observable demographics that correlate with the change in noise from the PBN implementation.

5 Results

5.1 Value-Added Model Results

I present my primary results from estimating the value-added model using OLS in Table 1. The table provides estimates for the impact of an additional loud flight per hour and of increasing the amount of flight noise during the school year by 1 dB on standardized test performance measured in standard deviations (SDs) above or below the annual mean test score. My estimates suggest a negative impact of additional loud noise events on math test scores, but not on ELA test scores. I estimate that an additional loud noise event per hour reduces standardized math test scores by 0.0030 SDs. Using the same model, I estimate that increasing school year flight noise by 1 dB reduces math test scores 0.0070 SDs.

Table 1: Impact of One Additional Loud Noise Event per Hour on Exam Performance

	Aircraft Leq		SEL Events	
	Math	ELA	Math	ELA
Estimate	-0.0070**	0.0011	-0.0030***	-0.0001
	(0.0027)	(0.0020)	(0.0011)	(0.0009)
Observations	57,617	57,596	57,617	57,596
R^2	0.75	0.69	0.75	0.69

*/**/*** denote significance at the $p < 0.1/0.05/0.01$ level.
Test performance is measured in standard deviations from the statewide mean score.
Standard errors are clustered at the school level.

The magnitude of the math estimates is large. Heissel et al. (2022) estimate that attending a school just downwind of a highway instead of one just upwind of a highway reduces standardized test scores by 0.04 SDs due to the large increase in exposure to air pollution.[13] Moving from a school in my dataset at the 25th percentile of exposure during the 2019 school year (47.6 dB) to the 75th (61.9 dB) is associated with a 0.10 SD reduction in score, more than twice as large as the impact of missing 10 school days during the year (Liu et al. 2021).[14]

[13]Based on their first stage, the effect of an additional microgram per cubic meter of $PM_{2.5}$ is 0.02 SDs, which is similar to the 0.016 that Gilraine and Zheng (2024) find.

[14]Goodman (2014) uses moderate snowfall as an instrument for absences and estimates a much larger impact of

The difference in my estimates of the impacts of noise on math and ELA is consistent with studies on the importance of working memory for cognitive task completion. Working memory is a key input for mathematical problem solving (Lin and Powell 2022) and distractions impede math performance (Vallée-Tourangeau et al. 2016). Further, the effects of working memory on math and language tasks are not equal. Problem solving draws on domain-specific working memory as well as general working memory. Visual-spatial memory loads draw more additional effort from general working memory, and visual-spatial tasks have greater attention demands than verbal ones (Unal et al. 2022). Visual-spatial working memory has a larger impact on visual-spatial task performance than verbal working memory does on verbal task performance (Giofrè et al. 2018). The relatively larger taxation of math tasks on working memory and importance of math working memory suggest that distractions during exams have the potential to impede math performance more than verbal performance.

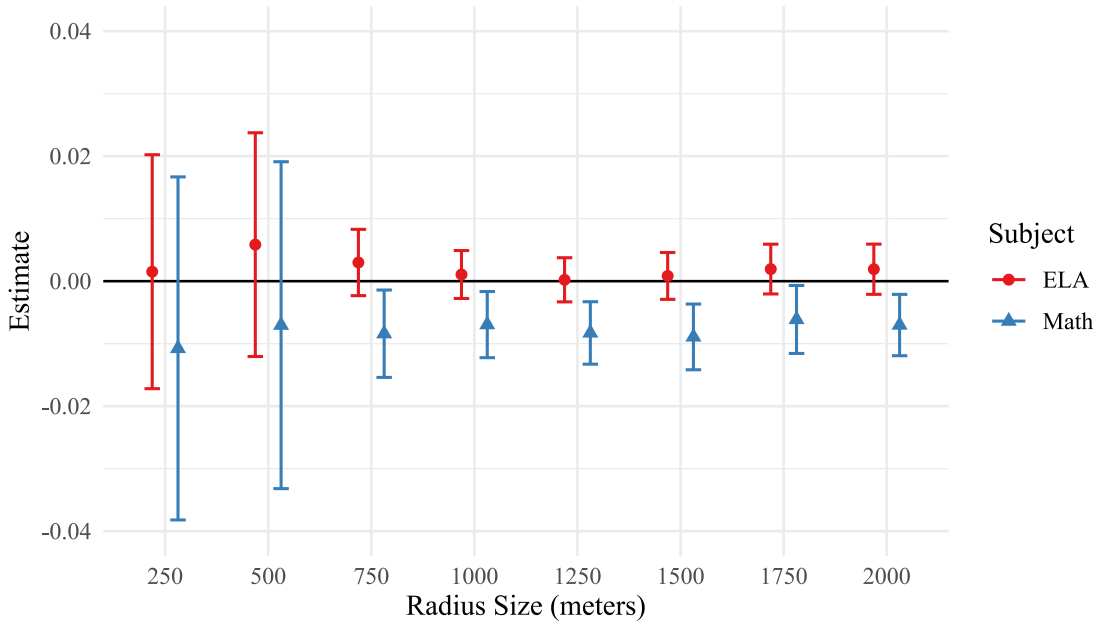
5.2 Sensitivity Analysis

I conduct several sensitivity analyses to determine whether my findings are robust to alternate specifications. First, I consider whether my choices of matching radius around noise monitors is driving my results. Expanding the radius around noise monitors increases the sample size, but reduces the accuracy of the noise pollution variable, potentially inducing more measurement error. Figure 4 shows that for most buffer distances considered from 250 to 2000 meters the estimate of the impact of noise pollution on math test scores is significant, and the magnitude is fairly consistent across all distances.

I also consider whether my choice to exclude the 2015 test year and include the 2014 test year drives my results. Under my preferred specification, I exclude the 2015 test year because the flight path change happens in April of 2015 (and therefore may impact the exam period) and I am missing data for January through April 2015. I run a regression including the 2015 tests where the measure of noise pollution is the average number of SEL events per school day during September through December 2014 for the observations that year to account for the missing January to April data. I also consider excluding the 2014 test year as schools had flexibility in choosing whether or not to administer the exam that year. In Table 2, I compare my primary specification to these two specifications. The results are similar across all specifications.

absences on math test scores (0.05 SDs per absence) than do Liu et al. (2021). However, he notes that if teacher absences increase during moderate snowfall days, then this estimate captures the effect of teacher absences as well as student absences.

Figure 4: Robustness to Alternate Noise Monitor Radius Choices



Bars indicate the 95% confidence interval for each estimate.

Table 2: Robustness to Alternate Year Specifications

	Math			ELA		
	(1)	(2)	(3)	(4)	(5)	(6)
Aircraft Leq	-0.0070**	-0.0066***	-0.0074***	0.0011	-0.0011	0.0013
	(0.0027)	(0.0020)	(0.0026)	(0.0020)	(0.0017)	(0.0023)
Includes 2014	X	X		X	X	
Includes 2015		X			X	
Observations	57,617	61,890	53,403	57,596	61,842	53,382
R^2	0.75	0.74	0.75	0.69	0.68	0.69

*/**/*** denote significance at the $p < 0.1/0.05/0.01$ level.
Test performance is measured in standard deviations from the statewide mean score.
Preferred model excludes 2015 test year due to change in flight paths in April 2015.
Standard errors are clustered at the school level.

5.3 Difference in Differences Results

My difference in differences model provides estimates that corroborate the results from my primary model. Table 3 provides results for both math and ELA outcomes using the change in flight Leq and the change in SEL events per hour as the continuous treatment variables. I also provide event study results in Online Appendix Figure 4 and Online Appendix Figure 5 to show that there are no pre-period trends that threaten the parallel trends assumption. The point estimates from the difference in differences model are significant at the $p < 0.05$ level for the Leq math estimate and $p < 0.1$ level for the SEL math events estimate. These estimate are larger than those from the value-added model I presented in Section 5.1. One potential reason for this is that some students might spend multiple years exposed to noise at these schools. The value-added model is focused on the effects of changes in flight noise during one school year, but the difference in differences estimates will capture the effect of multiple years for students who spend multiple pre- or post-period years attending a school. I prefer the value-added model results as they more cleanly capture the effect of additional noise during a school year.

Table 3: Difference in Differences Estimates of Impact on Test Performance

Independent variable:	Post × Δ Flight Leq		Post × Δ SEL Events per Hour	
Dependent variable:	Math	ELA	Math	ELA
Estimate	-0.0214**	-0.0012	-0.0077**	0.0002
	(0.0096)	(0.0118)	(0.0037)	(0.0035)
Observations	57,742	57,825	57,742	57,825
R^2	0.29	0.27	0.29	0.27

*/**/*** denote significance at the $p < 0.1/0.05/0.01$ level. Test performance is measured in standard deviations from the statewide mean score.
Standard errors are clustered at the school level.

6 Conclusion

This is the first study, to my knowledge, to provide causal estimates of the impact of noise pollution on standardized test performance. I use exogenous changes in aircraft noise to assess the impact of noise pollution on student performance. I find a meaningfully large impact of aircraft noise during the school year on math test scores but not on ELA test scores. Further, my estimates suggest that the magnitude of these results is large. Moving from a school in my dataset at the 25th percentile of exposure during the 2019 school year (47.6 dB) to the 75th (61.9 dB) is associated

with a 0.10 SD reduction in score, more than twice as large as the impact of missing 10 days of school during a year (Liu et al. 2021).

A limitation of my work is that I cannot disentangle what share of the effect is from cumulative impacts on reduced learning and what share is from contemporaneous impacts on exam days. Washington allows schools to set their own testing dates and times (within a range of a couple months), and the school districts I contacted have not retained this information for the period of my study. As a result, I cannot determine the amount of pollution during test times and would need to use the average amount during the testing period instead. Average noise during the testing period is highly correlated with noise during the rest of the school year. Therefore, I cannot separate the effects on learning and encoding information from the effects on test taking and information recall. Future research should address this issue by choosing a setting where test dates are documented and there is variation between cumulative noise pollution exposure and exposure during the test. One possibility would be to take the approach of Ebenstein et al. (2016) and measure day-of-test exposure controlling for prior exposure in a setting where there is sufficient daily variation. Many airports change flight paths depending on the direction of the wind (Hener 2022) or when runways are under construction or repair, so this should be possible in an aircraft noise pollution context.

A second limitation is that I cannot measure differences in impacts between daytime and nighttime noise. The hedonics literature and airport regulations tend to use DNL because loud noise is more irritating to residents when it interrupts sleep. The health literature suggests that loud noise overnight interrupts sleep and increases stress levels, leading to adverse cardiovascular health (Münzel et al. 2014; Saucy et al. 2020; Vienneau et al. 2022). In my setting, overnight noise leads to less sleep which could impact student performance. Daytime noise is likely to distract students and directly interfere with learning and test taking. Unfortunately, I cannot disentangle the relative effects of daytime and nighttime noise because the two are highly correlated in my setting.[15] Future work could look at runway construction, which can have different effects on daytime and nighttime flight paths.

My finding of a negative impact of noise pollution on academic performance suggests steps for reducing the impacts and improving understanding of noise pollution. First, in the school context, sound insulation may mitigate damages at schools exposed to considerable amounts of flight noise. SeaTac and the Port of Seattle have been proactive about reducing noise pollution – their noise pollution monitoring system is more thorough and has been around longer than similar networks at other major airports, and they've focused on methods to reduce noise pollution.

Additionally, the Port of Seattle and FAA have spent hundreds of millions of dollars funding sound insulation projects for houses to mitigate pollution, and recently allocated $70 million to

[15]The correlation between daytime and nighttime Leq at the noise monitor by year level is 0.93, so there isn't enough variation to identify differences in impacts of the two.

insulate nine schools (with $30 million set aside for six more).[16] This funding was focused on the Highline School District, which is closer to the airport than many of the schools treated with an increase in loud noise events following the PBN implementation. This suggests a role for expanding the funding to schools further from the airport, but underneath flight paths. Future research should investigate noise insulation as a mitigating factor on the impact of noise pollution on cognition and and learning. However, it is worth noting that my estimates are local average treatment effects that are identified off noise pollution variation at schools exposed to a considerable amount of flight noise. Therefore, I provide no evidence that aircraft noise pollution is likely to reduce achievement at areas with very little flight noise.[17]

In line with the previous recommendation, most airports do not currently have a robust noise monitoring system. The areas identified as affected by airport noise are based on 65 dB DNL measures and are therefore generally very close to airports. My findings suggest large impacts to schools further away (a cluster of schools that were impacted most negatively by the flight path changes was 8-10 miles – 13-16 kilometers – away from the airport). This suggests sound proofing and other assistance is focused too narrowly and should be expanded to loud areas further from the airport. Ensuring adequate data collection to measure impacts to communities further from the airport would allow for monitoring of the impacts of noise pollution for schools and individuals, and allow researchers to estimate a dose response curve. Such research is important for understanding where to target noise insulation funding.

[16]The FAA funds insulation for houses exposed to excessive noise pollution from air travel, and the Port of Seattle has used airport revenues to fund additional sound insulation as well.

[17]For example, moving from 10 to 11 dB of flight noise per school year may have no detectable impact on learning given the low amount of cumulative exposure.

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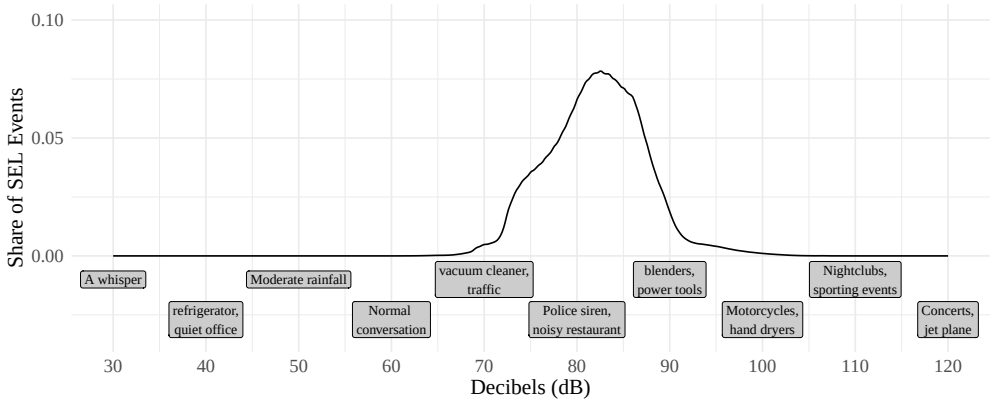
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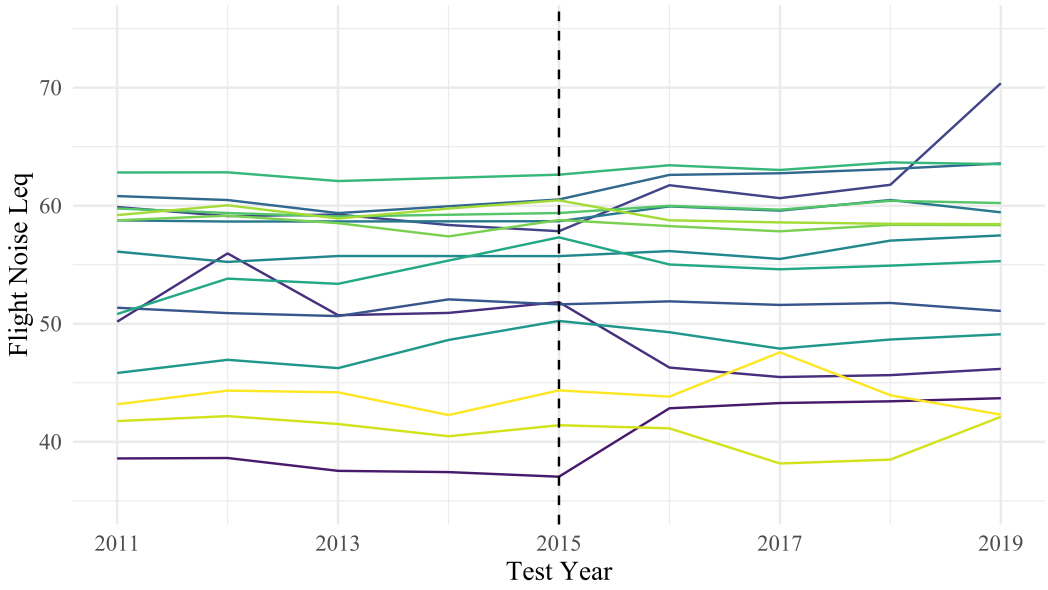
Online Appendix

Online Appendix Figure 1: Distribution of Sound Exposure Level Events Recorded by The Port of Seattle's Monitor System



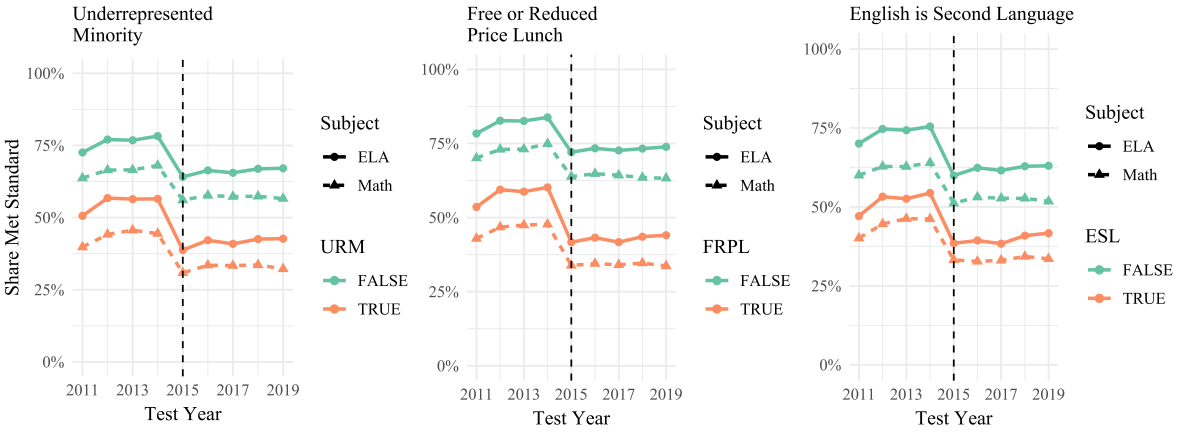
Notes: Monitor sample is limited to those that operated from 2009 through 2019. Density is for all sound exposure level events recorded by these monitors from 2009 through 2019.
Decibel level examples come from MDHearing.

Online Appendix Figure 2: Flight Noise Leq by Noise Monitor and School Year



Notes: Each line depicts the annual flight noise-specific Leq during the school year for a noise monitor.

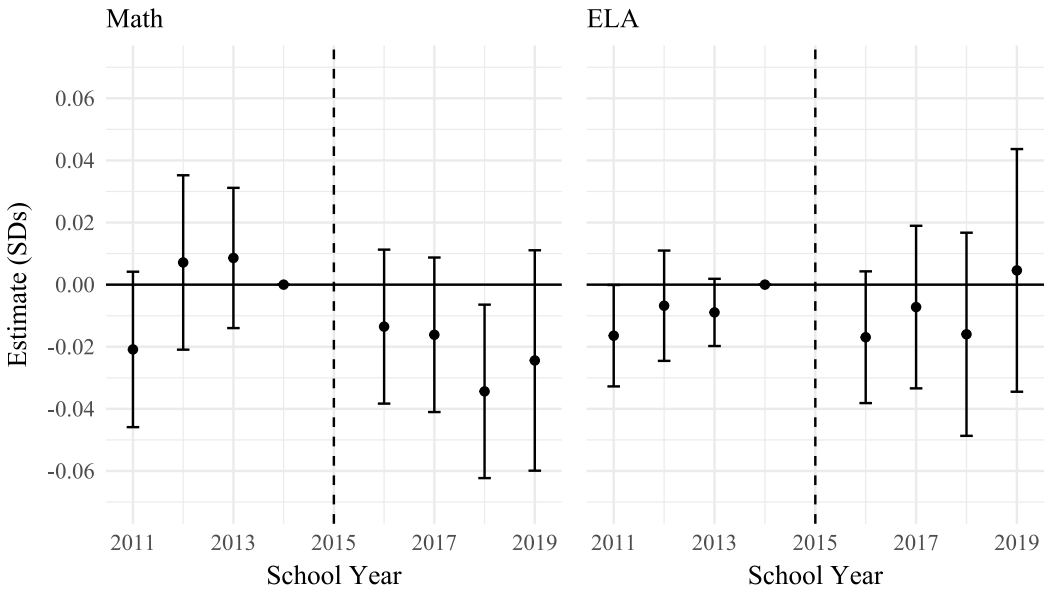
Online Appendix Figure 3: Impact of Test Regime Change by Demographic Group



Online Appendix Table 1: Difference in Differences Estimates of Impact on Demographics

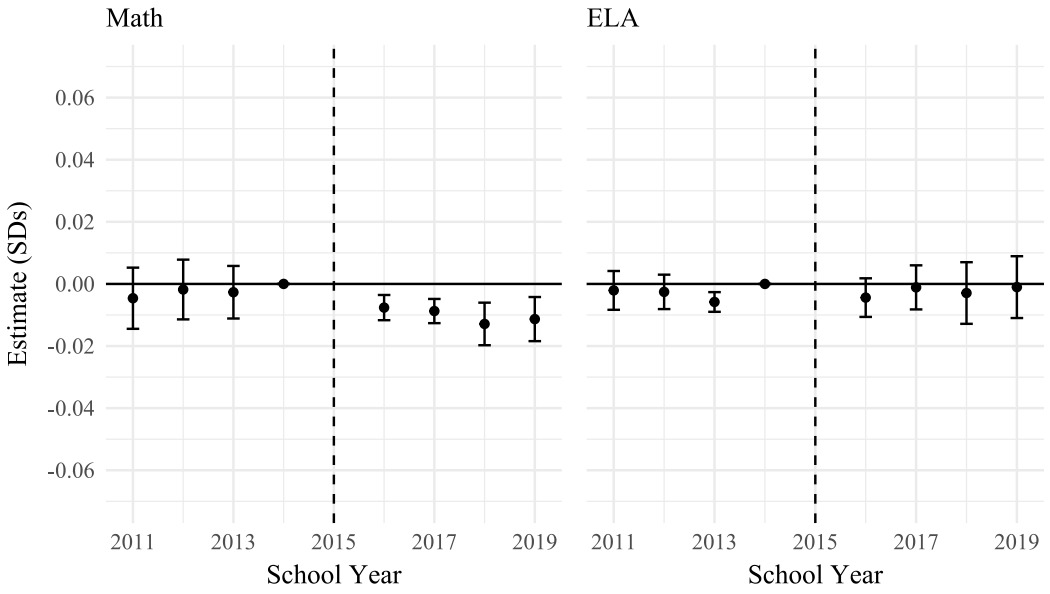
Independent variable:	Post × Δ Flight Leq			Post × Δ SEL Events per Hour		
Dependent variable:	FRPL	ESL	URM	FRPL	ESL	URM
Estimate	0.0036	0.0044	0.0067	-0.0011	-0.0046	0.0031
	(0.0058)	(0.0090)	(0.0080)	(0.0022)	(0.0029)	(0.0028)
Observations	57,927	57,927	53,831	57,927	57,927	53,831
R^2	0.15	0.08	0.07	0.15	0.08	0.07

*/**/*** denote significance at the $p < 0.1/0.05/0.01$ level. FRPL is an indicator variable for receipt of free or reduced price lunch. ESL is an indicator variable for whether English is the student's second language. URM is an indicator for whether the student is from an underrepresented minority racial/ethnic group.
Regressions include fixed effects for school-grade.
Standard errors are clustered at the school level.



Online Appendix Figure 4: Event Study Plot of Difference in Differences Model for Δ Flight Leq

Bars indicate the 95% confidence interval for each estimate.



Online Appendix Figure 5: Event Study Plot of Difference in Differences Model for Δ SEL Events

Bars indicate the 95% confidence interval for each estimate.